

24 . DIET RECOMMENDATION-DISEASE

AIM

To implement a simple Prolog expert system that suggests suitable foods based on the disease entered by the user.

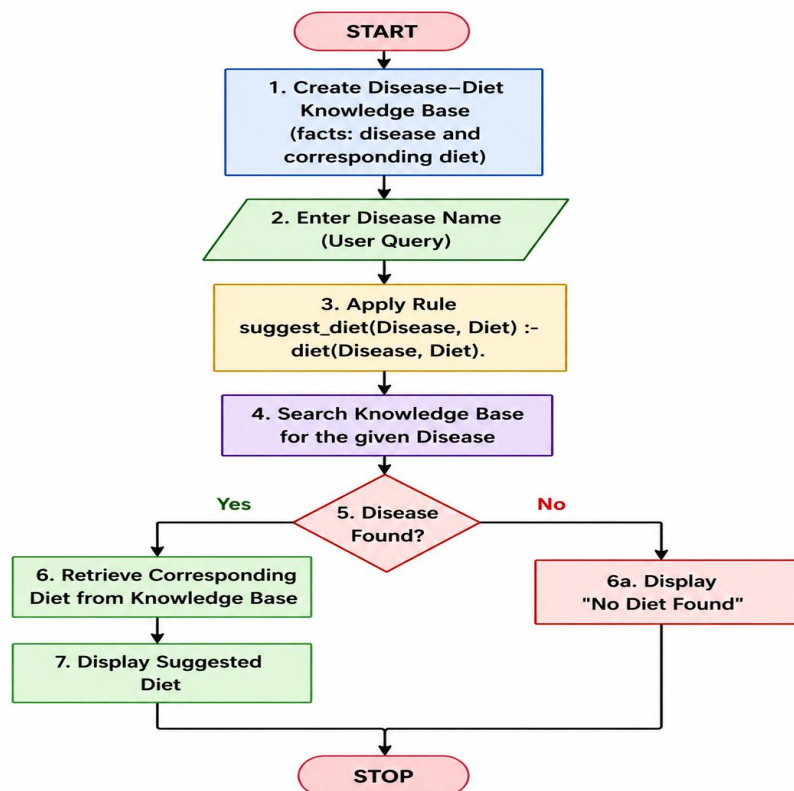
OBJECTIVE

To use Prolog facts and rules to provide diet suggestions for different diseases.

ALGORITHM

1. Define diseases and their recommended foods as facts.
2. Create a rule to connect a disease with its food.
3. Ask the user to enter a disease.
4. Search the knowledge base for the corresponding food.
5. Display the recommended food.

FLOWCHART



CODE

```
% Diet Suggestion System
```

```
diet(diabetes, oats).
```

```
diet(diabetes, vegetables).
```

```
diet(diabetes, beans).
```

```
diet(anemia, spinach).
```

```
diet(anemia, beetroot).
```

```
diet(anemia, lentils).
```

```
diet(hypertension, banana).
```

```
diet(hypertension, spinach).
```

```
diet(hypertension, oats).
```

```
suggest(Disease) :-
```

```
    diet(Disease, Food),
```

```
    write(Food), nl,
```

```
    fail.
```

```
suggest(_).
```

```
start :-
```

```
    write('Enter disease: '),
```

```
    read(Disease),
```

```
    write('Recommended foods:'), nl,
```

```
    suggest(Disease).
```

OUTPUT

```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File Edit Settings Run Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/Users/Admin/OneDrive/Documents/Diet Suggestion System.pl compiled 0.00 sec, 12 clauses
?- start.
Enter disease: amnesia.
Recommended foods:
true.

?- start.
Enter disease: anemia.
Recommended foods:
spinach
beetroot
lentils
true.

?- █
```

RESULT

The Prolog diet suggestion system was successfully implemented using simple facts and rules to suggest foods based on the given disease.